

Lecture 8 – Solar Photovoltaics

Part II – Circuit and Systems

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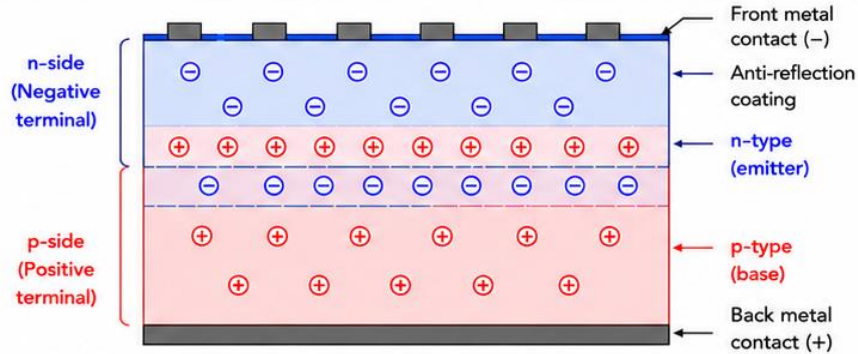


Outline

- P-N Junction Diodes
- Equivalent Circuits and I-V Curve
- Cell, Module, and Array
- Shading Impact and Mitigation
- Maximum Power Point Tracker

Polarity of a PV Cell

1. Structure of a PV Cell (p-n Junction) – Equilibrium (Dark)

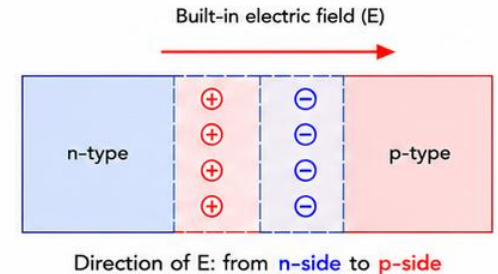


- ⊕ Positive charge (e.g., holes or ionized donor ions) ⊖ Negative charge (e.g., electrons or ionized acceptor ions)

2. Built-in Electric Field in the Depletion Region

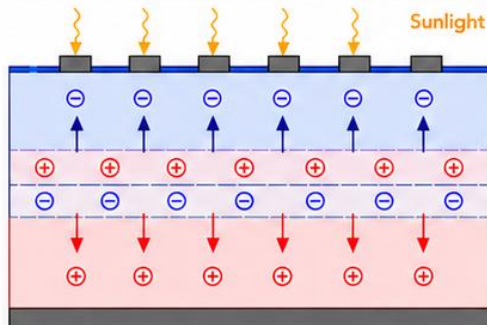
After formation of the p-n junction (equilibrium, dark):

- Electrons diffuse from n → p
- Holes diffuse from p → n
- Fixed ionized charges remain in the depletion region:
 - Positive ionized donor ions on n-side
 - Negative ionized acceptor ions on p-side
- This creates the built-in electric field.



- ⊕ Positive ionized donors (on n-side) ⊖ Negative ionized acceptors (on p-side)

3. Under Illumination (Photo-Generated Carriers)

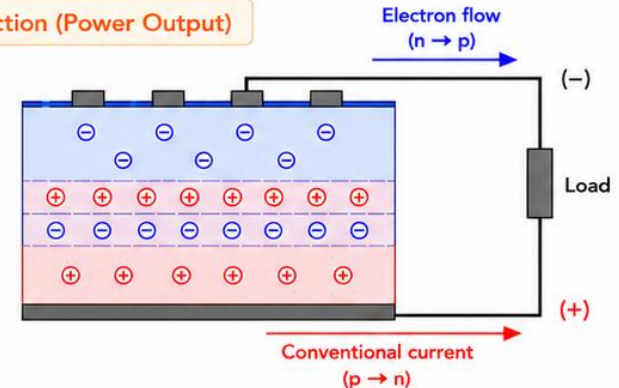


- Photons generate electron-hole pairs.
- Internal electric field (E) separates them:
 - Electrons (−) are pushed up to n-side
 - Holes (+) are pushed down to p-side
- Electrons accumulate on n-side (negative charge)
- Holes accumulate on p-side (positive charge)

4. External Circuit Connection (Power Output)

When connected to a load:

- Electrons flow through the external circuit from n-side to p-side.
- Conventional current flows from p-side (+) to n-side (−).



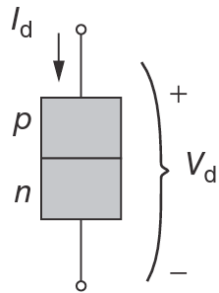
Summary

- The built-in electric field points from n-side to p-side.
- Under illumination, electrons (negative) move to n-side and holes (positive) move to p-side.
- Therefore, the p-side becomes positive and the n-side becomes negative.
- The external voltage polarity is: p-side positive (+), n-side negative (−).

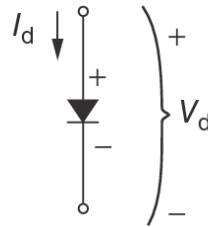
Terminal Polarity (Standard Silicon PV Cell)

- p-side (base) → Positive (+)
n-side (emitter) → Negative (−)

P-N Junction: Forward Bias vs Reverse Bias



(a) p - n junction diode



(b) Symbol for real diode

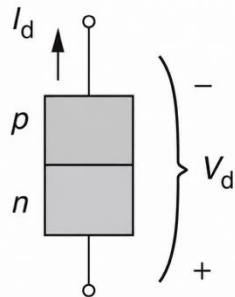
Forward current is dominated by majority carrier diffusion.

When **forward bias** is applied (p-side connected to positive terminal; n-side connected to negative terminal):

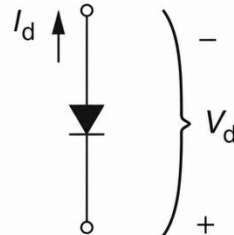
- the barrier becomes smaller
- majority carriers can more easily diffuse across the junction

As a result:

- diffusion current increases dramatically
- large diode current flows



(a) p - n junction reverse bias



(b) Symbol for real diode (reverse bias)

Reverse current is dominated by minority carrier drift due to the depletion-region electric field.

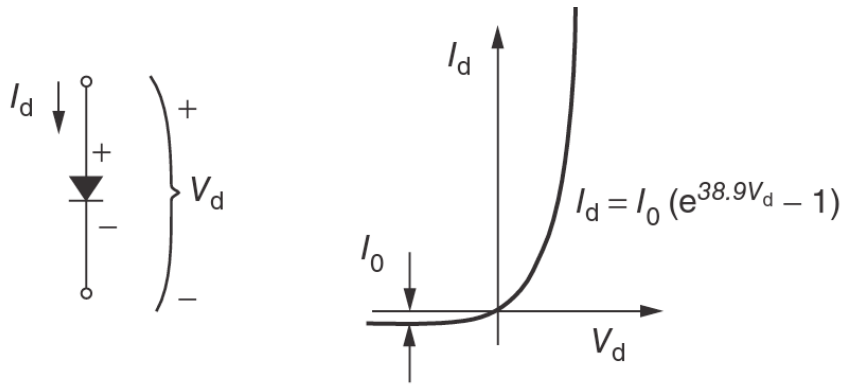
When **reverse bias** is applied (p-side connected to negative terminal; n-side connected to positive terminal):

- increases the potential barrier
- widens the depletion region
- strengthens the internal electric field

As a result:

- majority carriers are pulled away from the junction
- majority-carrier diffusion across the junction becomes extremely small

P-N Junction: Shockley Diode Equation



(b) Symbol for real diode

(c) Diode characteristic curve

Shockley diode equation:

$$I_d = I_0 (e^{qV_d/kT} - 1) \quad (5.3)$$

where I_d is the diode current in the direction of the arrow (A), V_d is the voltage across the diode terminals from the p-side to the n-side (V), I_0 is the reverse saturation current (A), q is the electron charge (1.602×10^{-19} C), k is Boltzmann's constant (1.381×10^{-23} J/K), and T is the junction temperature (K).

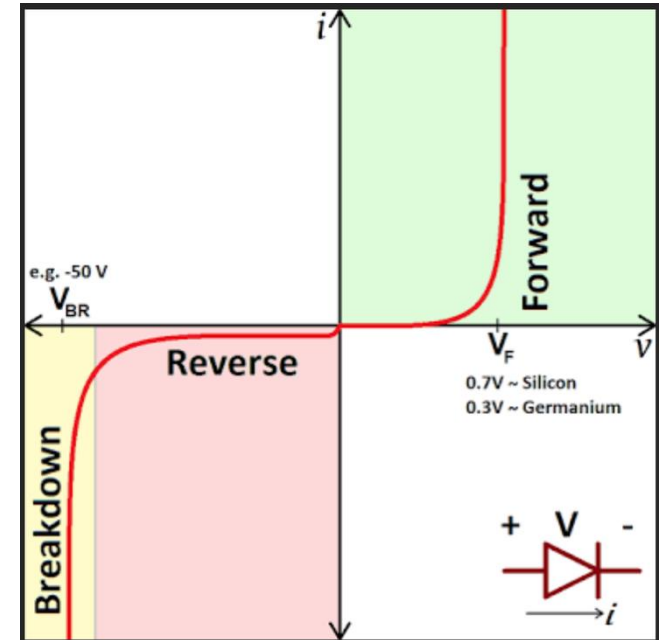
$$\frac{qV_d}{kT} = \frac{1.602 \times 10^{-19}}{1.381 \times 10^{-23}} \cdot \frac{V_d}{T \text{ (K)}} = 11,600 \frac{V_d}{T \text{ (K)}}$$

$$\Rightarrow I_d = I_0 (e^{38.9V_d} - 1) \quad (\text{at } 25^\circ\text{C}) \quad (5.5)$$

Reverse Saturation Current I_o

Reverse saturation current I_o :

- It is the small current flowing through a p-n junction under reverse bias.
- Mainly caused by **thermally generated minority carriers** (electrons in the p-type region and holes in the n-type region)
- I_o already exists internally at equilibrium as the minority-carrier **drift current**
- At equilibrium it is exactly balanced by **diffusion current**, the external current is zero

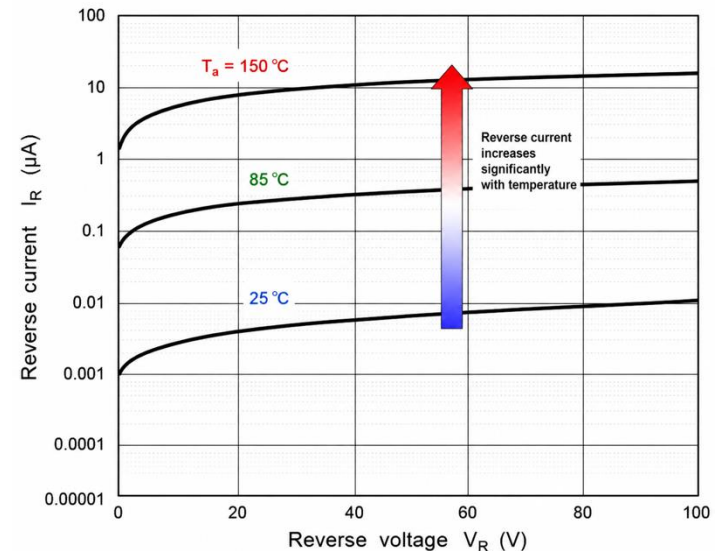


- Under reverse bias: the depletion region widens the electric field sweeps minority carriers across the junction
- It is called “saturation” current because: increasing reverse voltage (up to the breakdown voltage) does not significantly increase the carrier supply
- It is usually very small: $10^{-12} \sim 10^{-9}$ A, for typical silicon diodes.

Reverse Saturation Current I_o (cont'd)

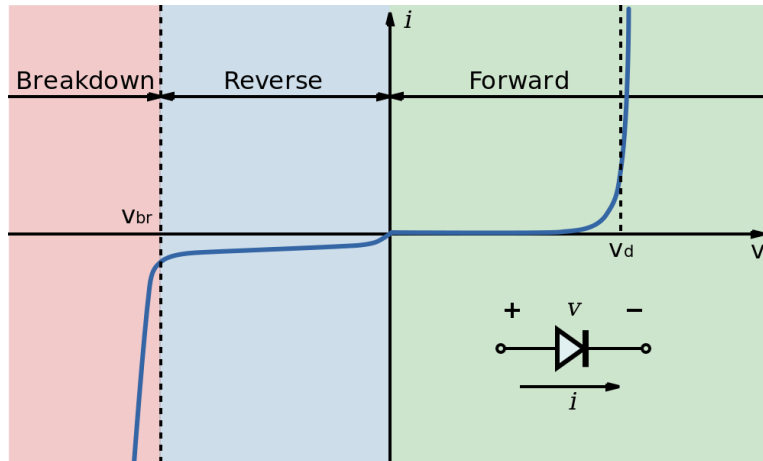
- I_o increases with temperature: Approximately doubles for every 10° C rise

$$I_{o2} = I_{o1} \times 2^{\frac{T_2 - T_1}{10}}$$



- In photovoltaic cells: larger I_o means:
 - more carrier recombination
 - more leakage through the junction
 - worse semiconductor quality
 - more internally “lost” current
- Smaller I_o leads to higher open-circuit voltage and better efficiency.
- I_o can be reduced by improving semiconductor quality, minimizing carrier recombination and leakage through
 - ✓ surface passivation
 - ✓ optimized doping
 - ✓ operating the PV cell at lower temperatures

Breakdown Voltage



Device Type	Typical Reverse Breakdown Voltage
Silicon PV cell (single cell)	5–20 V
Crystalline silicon PV cell	10–15 V
Small-signal silicon diode	50–100 V
Power rectifier diode	100–1600 V
Zener diode	2–200 V
Schottky diode	20–200 V

Breakdown voltage is the reverse-bias voltage at which a diode suddenly begins conducting a very large reverse current.

If the reverse voltage becomes too large:

- built-in electric field becomes extremely strong
- carriers are rapidly generated
- reverse current increases dramatically

For PV cells: reverse breakdown is important because shaded solar cells can become reverse biased, causing:

- hot spots
- overheating
- permanent damage

Bypass diodes are used to limit reverse voltage and protect PV cells.

Electrical Characteristics of PV Cells

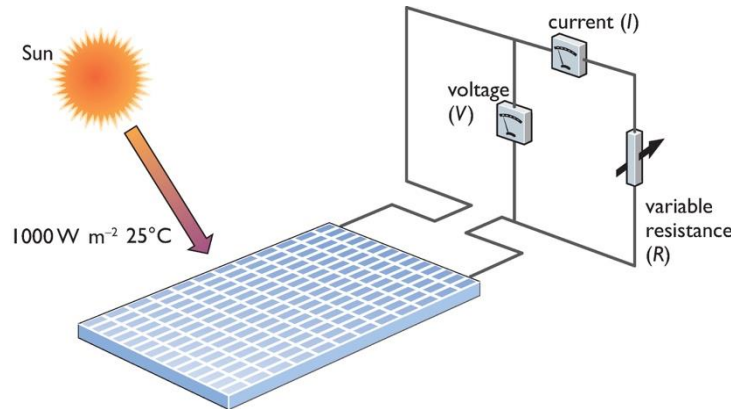
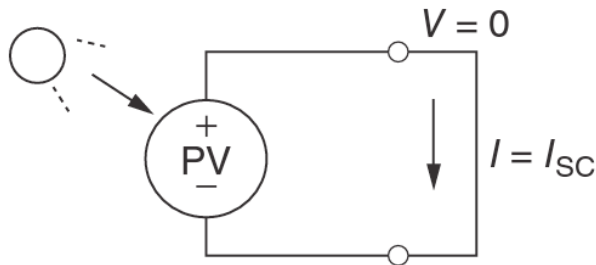
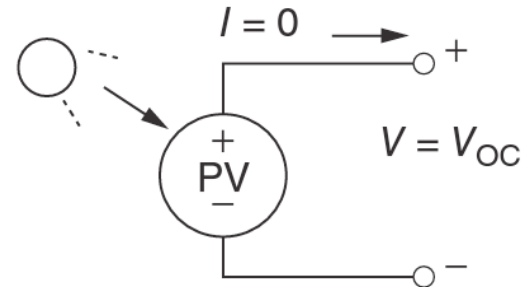


Figure: A PV cell connected to a variable resistance, with ammeter and voltmeter



(a) Short-circuit current



(b) Open-circuit voltage

$$R = 0$$

Short circuit current I_{sc} :

get maximum current in the circuit,
which is proportional to solar intensity

$$R \rightarrow \infty$$

Open circuit voltage V_{oc} :

get maximum voltage across the cell

PV Cell: I_{sc} & V_{oc}

- I_{sc} is proportional to solar intensity because stronger sunlight delivers more photons to the PV cell, generating more electron-hole pairs per second, which directly increases the photocurrent and therefore the short-circuit current.
- The bandgap energy E_g (expressed in eV) sets the theoretical upper bound on the open-circuit voltage V_{oc} (expressed in volts):
because one electron-hole pair generated by one photon cannot produce more electrical energy than the semiconductor bandgap.

$$V_{oc} \lesssim \frac{E_g}{q}$$

where q is the elementary charge (charge of one electron):

$$q = 1.602 \times 10^{-19} \text{ C}$$

For silicon $E_g \approx 1.12 \text{ eV}$, $V_{oc} < 1.12 \text{ V}$. But real silicon cells are typically: 0.6–0.75 V. This is because of:

- recombination losses
- entropy losses
- nonradiative processes
- resistive effects

Example

Example 5.2 A p–n Junction Diode. Consider a p–n junction diode at 25°C with a reverse saturation current of 10^{-9} A. Find the voltage drop across the diode when it is carrying the following:

- a. no current (open-circuit voltage)
- b. 1 A
- c. 10 A

Solution

- a. In the open-circuit condition, $I_d = 0$, so from Equation 5.5 $V_d = 0$.
- b. With $I_d = 1$ A, we can find V_d by rearranging Equation 5.5:

$$V_d = \frac{1}{38.9} \ln \left(\frac{I_d}{I_0} + 1 \right) = \frac{1}{38.9} \ln \left(\frac{1}{10^{-9}} + 1 \right) = 0.532 \text{ V}$$

- c. With $I_d = 10$ A,

$$V_d = \frac{1}{38.9} \ln \left(\frac{10}{10^{-9}} + 1 \right) = 0.592 \text{ V}$$

Equivalent Circuits for PV Cells

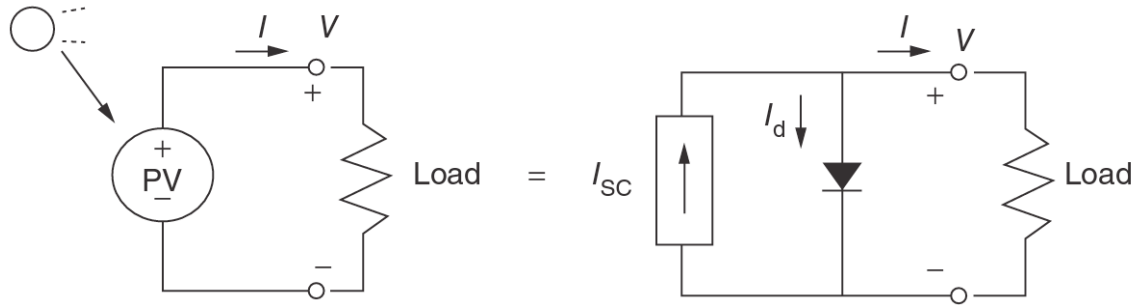


FIGURE 5.23 A simple equivalent circuit for a PV cell consists of a current source driven by sunlight in parallel with a real diode.

By KCL: $I = I_{SC} - I_d = I_{SC} - I_0 (e^{qV/kT} - 1)$

Set $I = 0$ $\Rightarrow$ $V_{OC} = \frac{kT}{q} \ln \left(\frac{I_{SC}}{I_0} + 1 \right)$ (5.9)

If $T = 25^\circ\text{C} = 298.15^\circ\text{K}$

$$I = I_{SC} - I_0 (e^{38.9V} - 1) \quad (5.10)$$

$$V_{OC} = 0.0257 \ln \left(\frac{I_{SC}}{I_0} + 1 \right) \quad (5.11)$$

Example

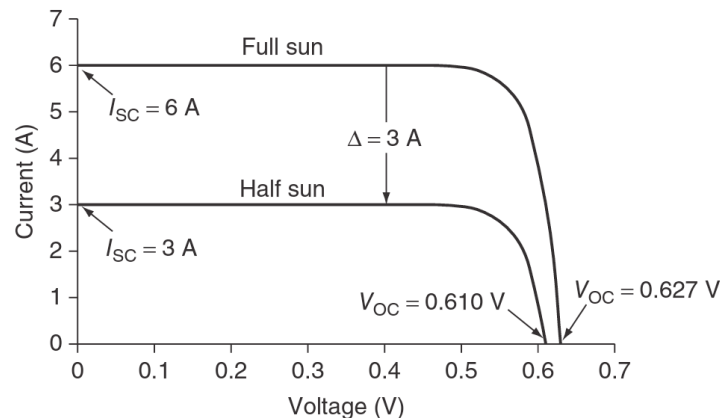
Example 5.3 The I – V Curve for a PV Cell. Consider a 150 cm^2 PV cell with reverse saturation current $I_0 = 10^{-12} \text{ A/cm}^2$. In full sun, it produces a short-circuit current of 40 mA/cm^2 at 25°C . What would be the short-circuit current and open-circuit voltage in full sun and again for 50% sun. Plot the resulting I – V curves.

Solution. The reverse saturation current $I_0 = 10^{-12} \text{ A/cm}^2 \times 150 \text{ cm}^2 = 1.5 \times 10^{-10} \text{ A}$. At full sun short-circuit current, I_{SC} , is $0.040 \text{ A/cm}^2 \times 150 \text{ cm}^2 = 6.0 \text{ A}$. From Equation 5.11, the open-circuit voltage for a single cell is

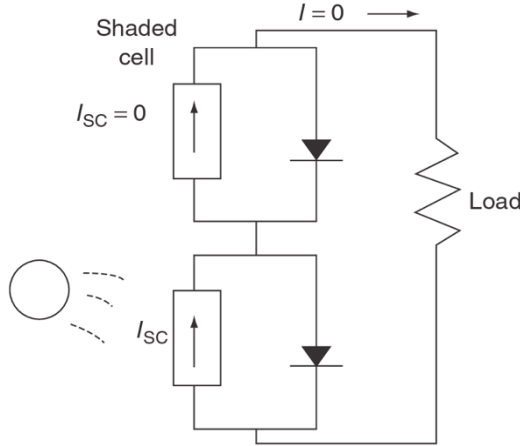
$$V_{\text{OC}} = 0.0257 \ln \left(\frac{I_{\text{SC}}}{I_0} + 1 \right) = 0.0257 \ln \left(\frac{6.0}{1.5 \times 10^{-10}} + 1 \right) = 0.627 \text{ V}$$

Since short-circuit current is proportional to solar intensity, at half sun $I_{\text{SC}} = 3 \text{ A}$ and the open-circuit voltage is

$$V_{\text{OC}} = 0.0257 \ln \left(\frac{3.0}{1.5 \times 10^{-10}} + 1 \right) = 0.610 \text{ V}$$



More Accurate Equivalent Circuits for PV Cells



- No current through the load if one PV cell is shaded.
- The real situation is not that bad.
- **More accurate model is needed!**

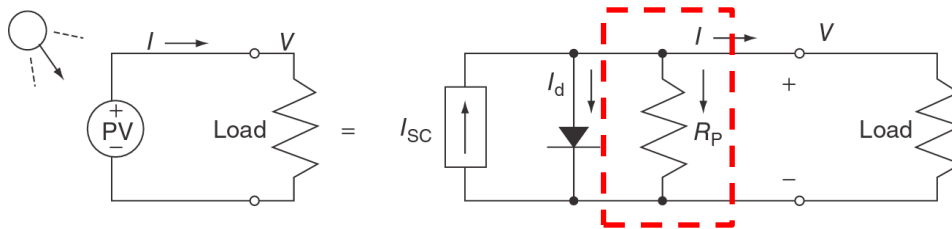
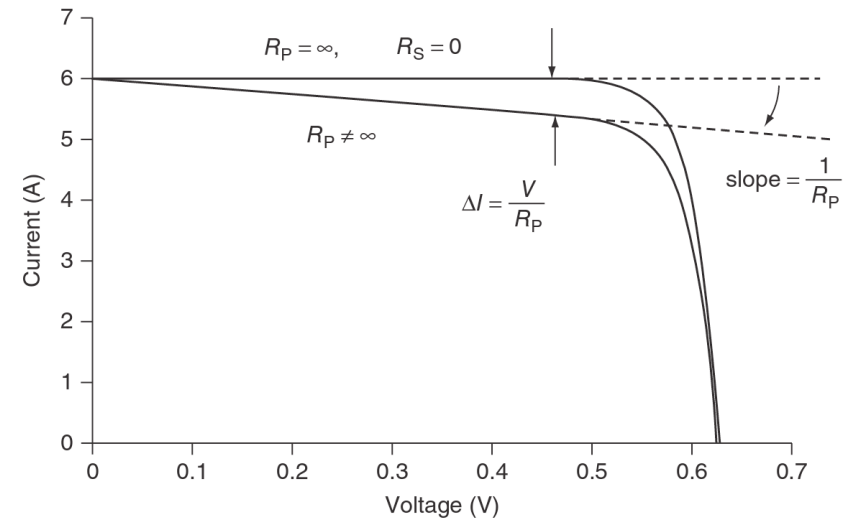


FIGURE 5.27 The simple PV equivalent circuit with an added parallel resistance.

$$I = (I_{SC} - I_d) - \frac{V}{R_P}$$

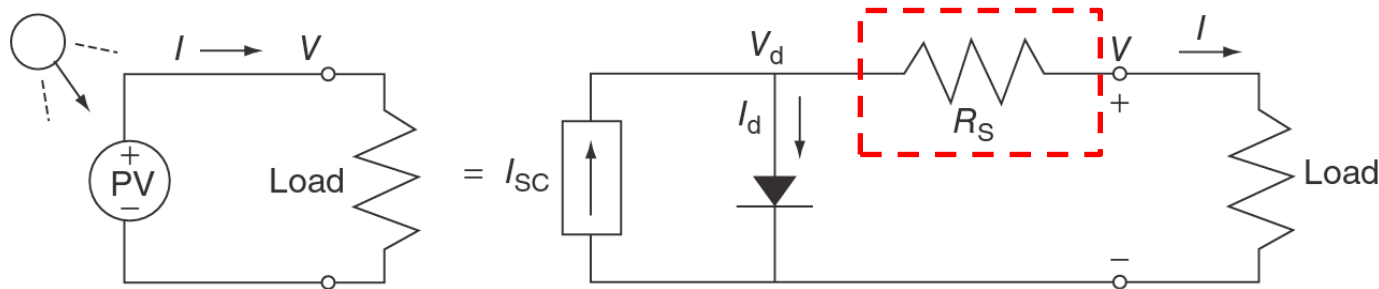
$$R_P > \frac{100V_{OC}}{I_{SC}}$$



A PV Equivalent Circuit with Series Resistance

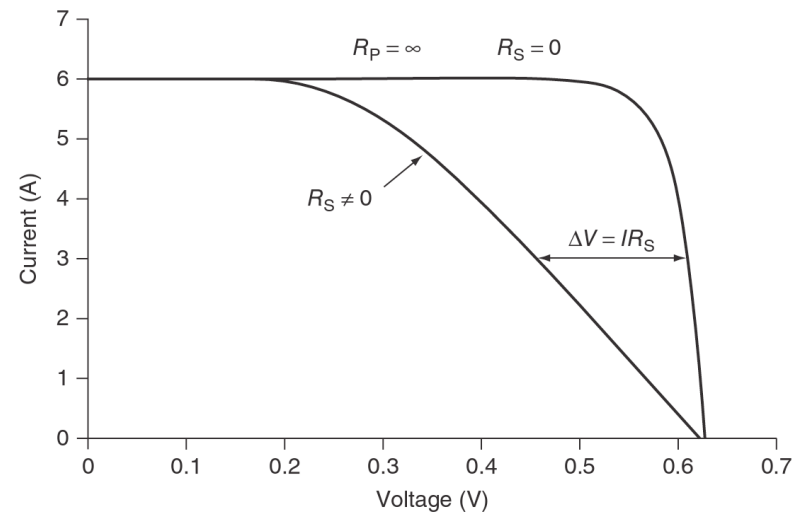
To model

- contact resistance associated with the bond between the cell and its wire leads.
- resistance of the semiconductor itself.

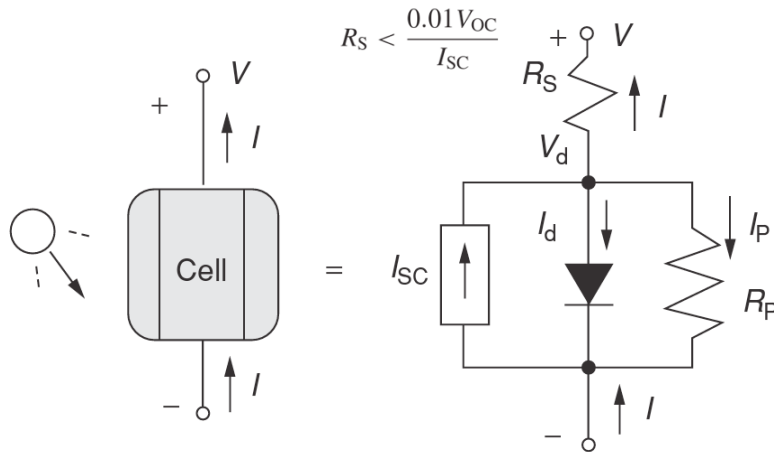


$$\begin{cases} I = I_{sc} - I_d = I_{sc} - I_0 (e^{qV_d/kT} - 1) \\ V_d = V + IR_s \end{cases}$$

$$\Rightarrow I = I_{sc} - I_0 \left\{ \exp \left[\frac{q}{kT} (V + IR_s) \right] - 1 \right\}$$



Equivalent Circuit with Series/Parallel Resistances



By KCL & KVL:

$$\begin{cases} I_{sc} = I + I_d + I_p \\ V_d = V + IR_S \end{cases}$$



$$I = I_{sc} - I_0 \left\{ \exp \left[\frac{q}{kT} (V + IR_S) \right] - 1 \right\} - \left(\frac{V + IR_S}{R_P} \right)$$

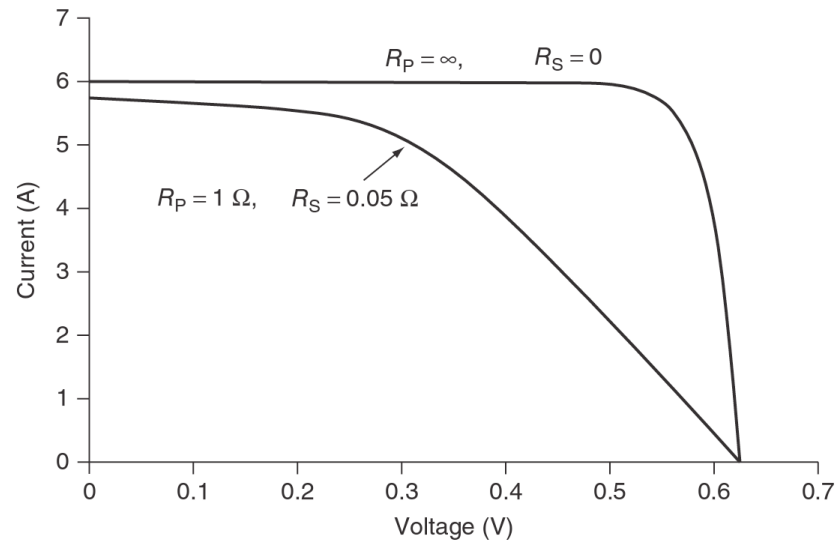
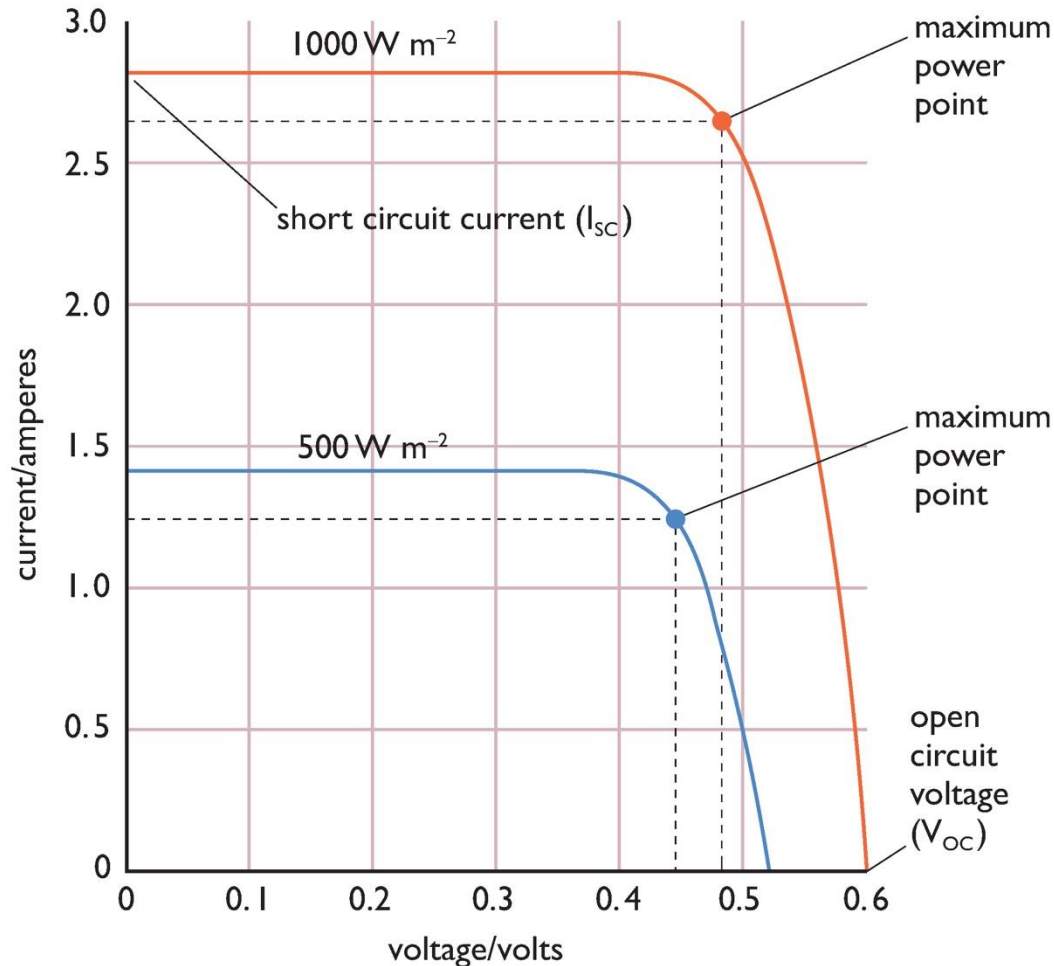


FIGURE 5.32 Series and parallel resistances in the PV equivalent circuit decrease both voltage and current delivered. To improve cell performance, high R_p and low R_s are needed.

Maximum Power Point



$I_{sc} \propto$ intensity of solar radiation

V_{oc} only changes slightly

We want the PV module to work at the **maximum power point (MPP)**.

Many PV systems incorporate MPP tracking circuit/control algorithms.

From Cell to Module & Array

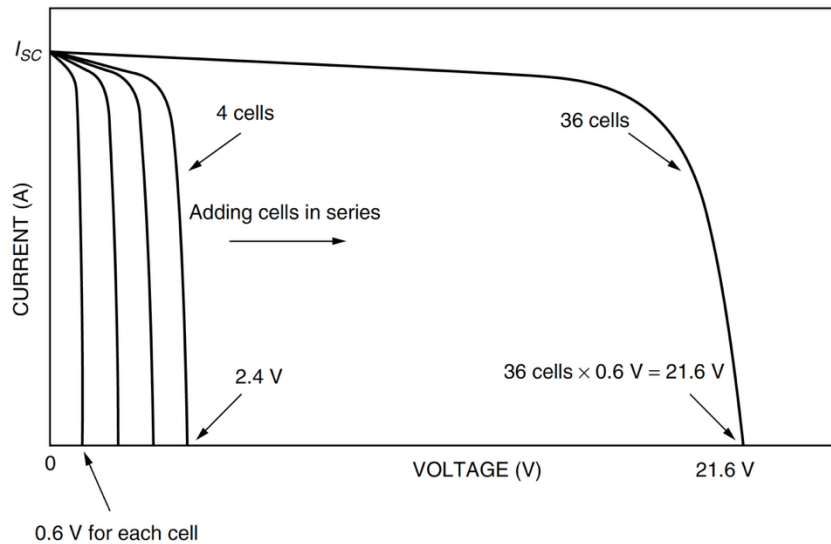
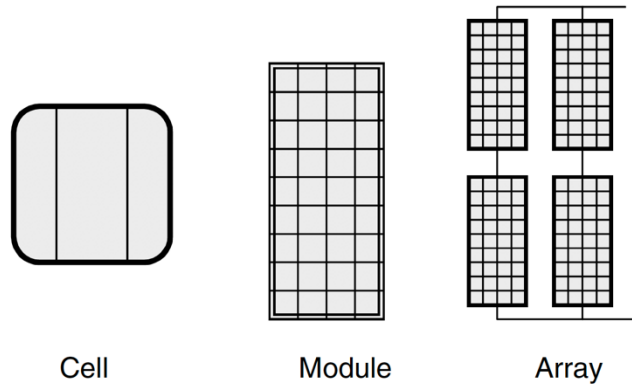


Fig-a: For cells wired in series, their voltages at any given current add: $V_{\text{module}} = n(V_d - IR_S)$

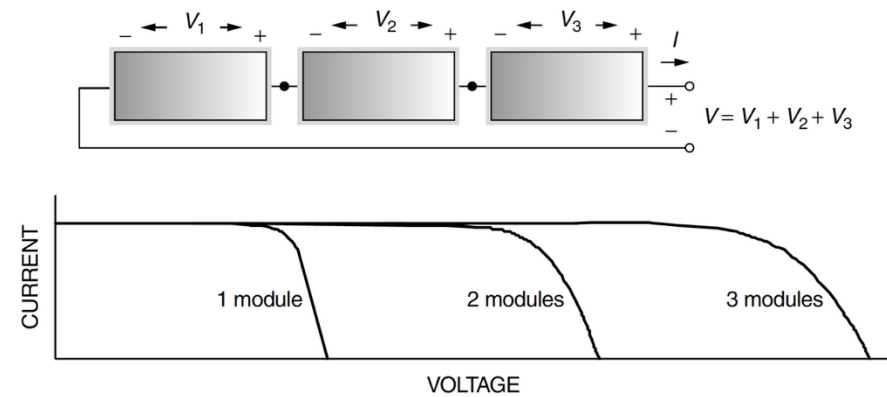


Fig-b: For modules in series, at any given current the voltages add.

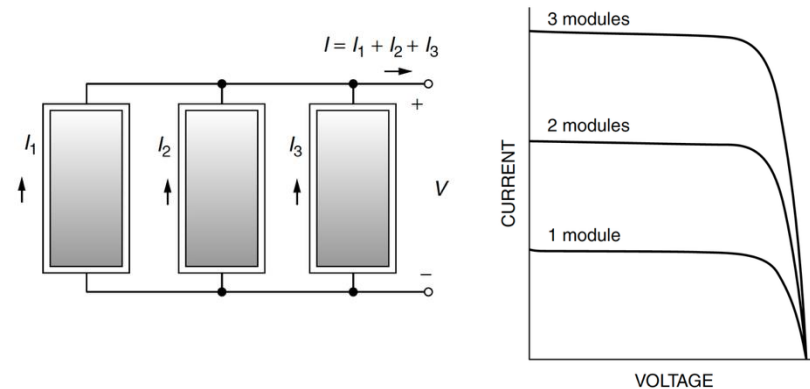
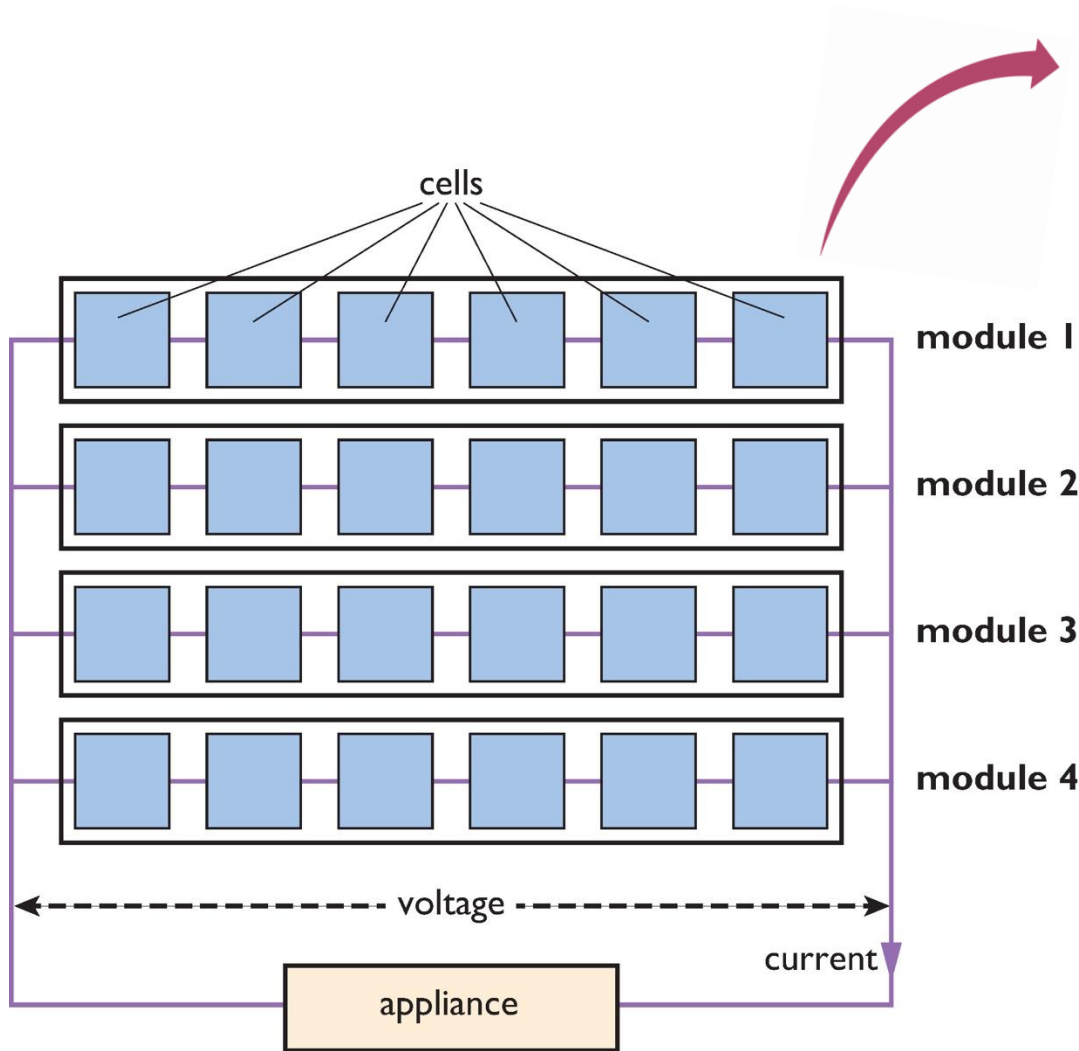


Fig-c: For modules in parallel, at any given voltage the currents add.

From Cell to Module and Array

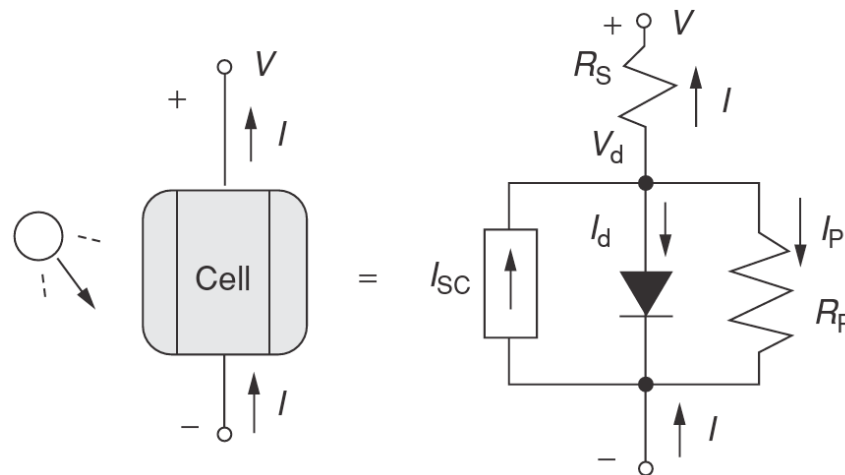


- $V_{\text{tot}} = \text{\# of cells} \times V_{\text{cell}}$
- $I_{\text{tot}} = \text{\# of modules} \times I_{\text{module}}$

Example: I-V relationship of PV Module

Example 5.4 Voltage and Current From a PV Module. A PV module is made up of 72 identical cells, all wired in series. With 1-sun insolation (1 kW/m^2), each cell has short-circuit current $I_{SC} = 6.0 \text{ A}$ and at 25°C , its reverse saturation current is $I_0 = 5 \times 10^{-11} \text{ A}$. Parallel resistance $R_P = 10.0 \Omega$ and series resistance $R_S = 0.001 \Omega$.

- Find the voltage, current, and power delivered when the diode voltage in the equivalent circuit for each cell is 0.57 V .
- Set up a spreadsheet for I and V of the entire module and present a few lines of output show how it works.



Example Solution

Solution

a. Using $V_d = 0.57$ V in Equation 5.19 along with the other data gives current

$$I = I_{SC} - I_0 (e^{38.9V_d} - 1) - \frac{V_d}{R_P}$$

$$= 6.0 - 5 \times 10^{-11} (e^{38.9 \times 0.57} - 1) - \frac{0.57}{10.0} = 5.73 \text{ A}$$

Under these conditions, Equation 5.21 gives the voltage produced by the 72-cell module:

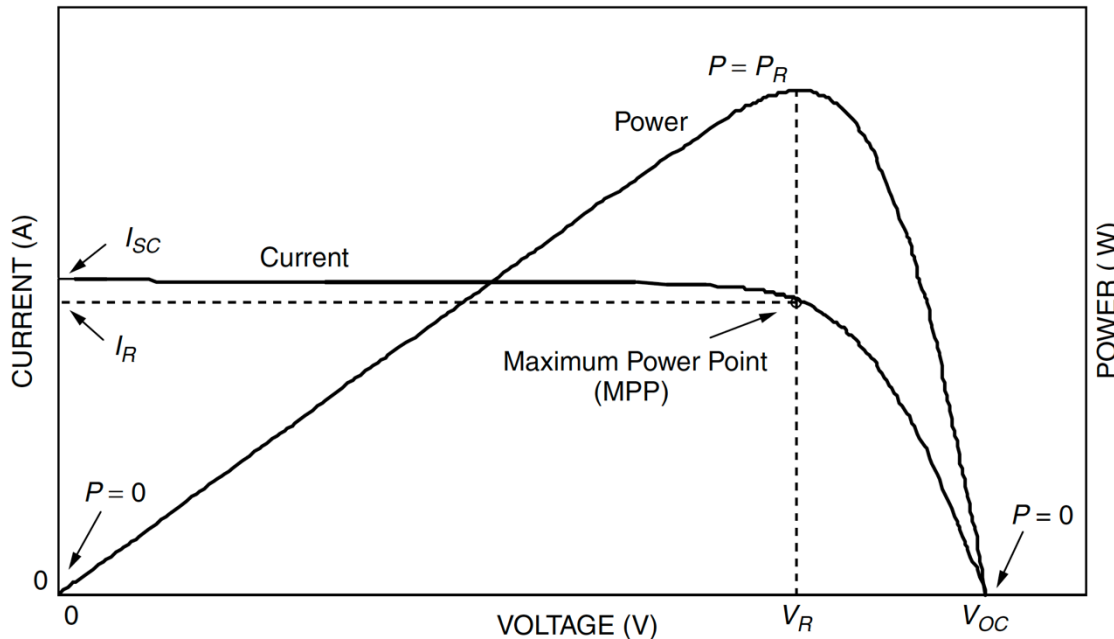
$$V_{\text{module}} = n (V_d - IR_S) = 72 (0.57 - 5.73 \times 0.001) = 40.63 \text{ V}$$

Power delivered is therefore

$$P(\text{Watts}) = V_{\text{module}} I = 40.63 \text{ V} \times 5.73 \text{ A} = 232.8 \text{ W}$$

Number of cells, n	72		
Parallel resistance/cell R_P (ohms)	10.0		
Series resistance/cell R_S (ohms)	0.001		
Reverse saturation current I_0 (A)	5E-11		
Short-circuit current at 1-sun (A)	6.0		
V_d	$I = I_{SC} - I_0 (e^{38.9V_d} - 1) - V/R_P$	$V_{\text{module}} = n (V_d - IR_S)$	$P(W) = V \times I$
0.53	5.9020	37.735	222.7
0.54	5.8797	38.457	226.1
0.55	5.8471	39.179	229.1
0.56	5.7996	39.902	231.4
0.57	5.7299	40.627	232.8
0.58	5.6276	41.355	232.7
0.59	5.4771	42.086	230.5
0.60	5.2555	42.822	225.0

Fill Factor (FF)



FF:

- around 70–75% for crystalline silicon solar modules
- closer to 50–60% for multi-junction amorphous-Si modules

$$\text{Fill factor (FF)} = \frac{\text{Power at the maximum power point}}{V_{OC} I_{SC}} = \frac{V_R I_R}{V_{OC} I_{SC}}$$

Physically, FF tells us:

- How close the PV cell operates to an ideal rectangular power source
- How much of the theoretical $V_{OC}I_{SC}$ power can actually be extracted
- The quality of the PV cell and its internal losses

Maximum Power Point

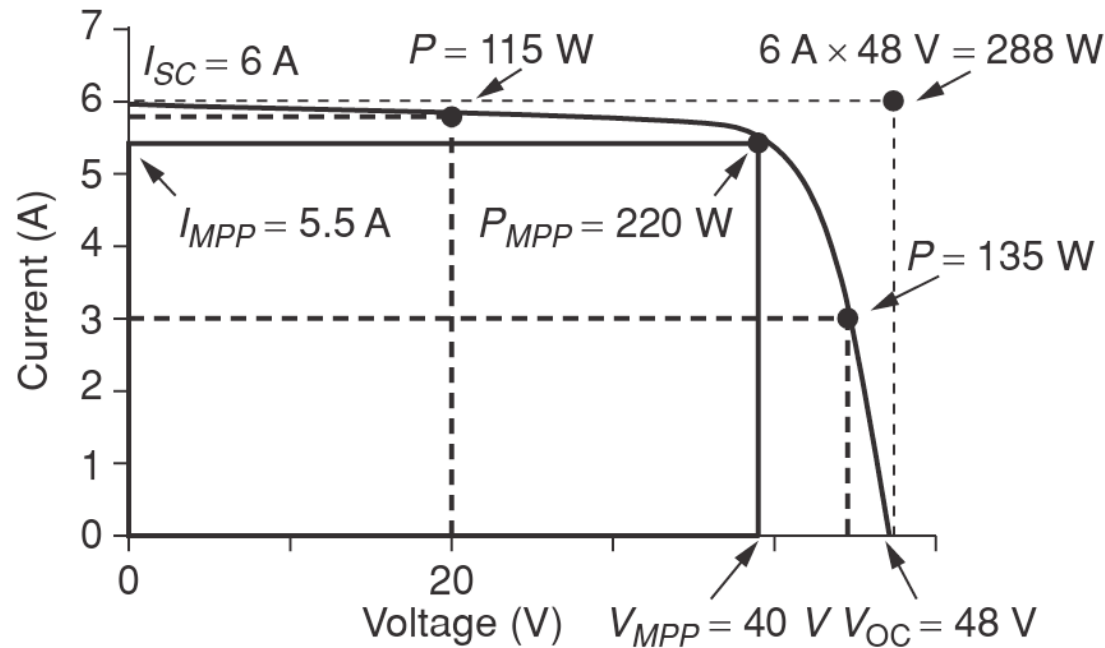


FIGURE 5.41 The maximum power point (MPP) corresponds to the biggest rectangle that can fit beneath the I - V curve. The fill factor (FF) is the ratio of the area (power) at MPP to the area formed by a rectangle with sides V_{OC} and I_{SC} .

$$\text{Fill factor} = \frac{40\text{V} \times 5.5\text{A}}{48\text{V} \times 6\text{A}} = \frac{220\text{W}}{288\text{W}} = 0.76$$

Improve Efficiency

PV cell/panel efficiency is:

$$\eta = \frac{P_{max}}{P_{in}} = \frac{V_{OC}I_{SC}FF}{P_{in}}$$

So, increasing I_{SC} , V_{OC} , and fill factor (FF) help improve efficiency.

However, there is an important tradeoff:

- Smaller bandgap materials absorb more sunlight $\rightarrow$ larger I_{SC}
- Larger bandgap materials produce higher voltage $\rightarrow$ larger V_{OC}

Modern high-efficiency PV technologies try to improve both simultaneously by:

- reducing recombination losses
- improving carrier collection
- lowering I_0
- using tandem/multi-junction structures
- improving optical absorption and passivation (coating the semiconductor surface with a thin insulating layer that neutralizes surface defects)

Impacts of Temperature & Insolation

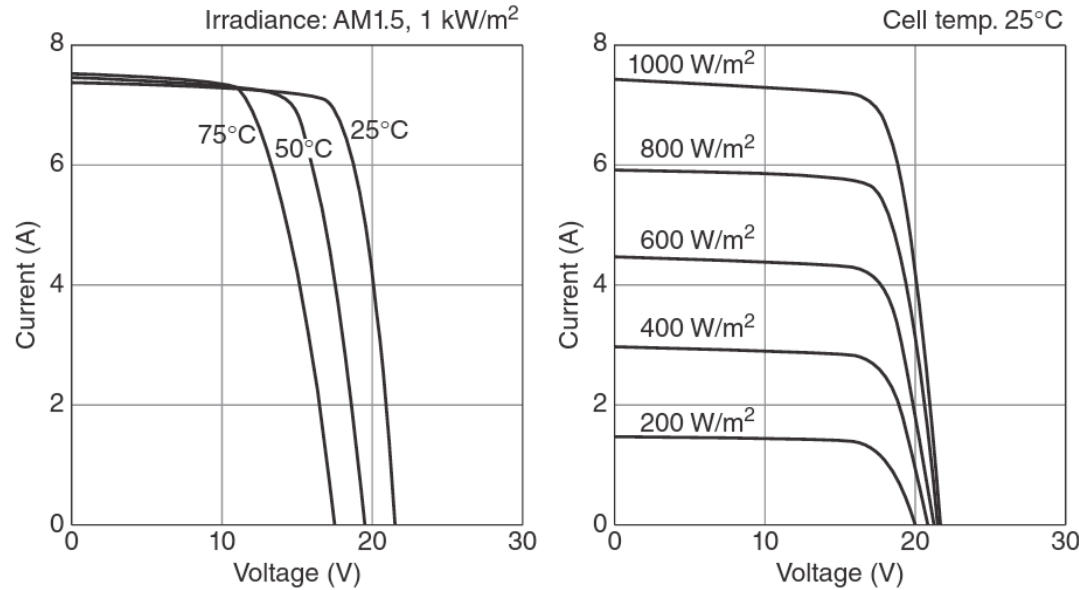


Fig: I-V curves under various cell temperatures and irradiance levels for the Kyocera KC120-1 PV module.

$$T_{\text{cell}} = T_{\text{amb}} + \left(\frac{\text{NOCT} - 20^\circ}{0.8} \right) \cdot S \quad \text{NOCT: nominal operating cell temperature}$$

where T_{cell} is cell temperature (°C), T_{amb} is ambient temperature, and S is solar insolation (kW/m²).

NOCT

- NOCT tells us: “Under standard outdoor operating conditions, how much hotter than ambient does the PV module become?”
- NOCT is typically around 40°C to 48°C for commercial modules

Typical example:

Parameter	Typical Value
Ambient temp during NOCT test	20°C
Irradiance	800 W/m ²
Wind speed	1 m/s
Typical NOCT	45°C

So if NOCT = 45°C:

$$T_{\text{cell}} = T_{\text{amb}} + \frac{45 - 20}{0.8} S$$

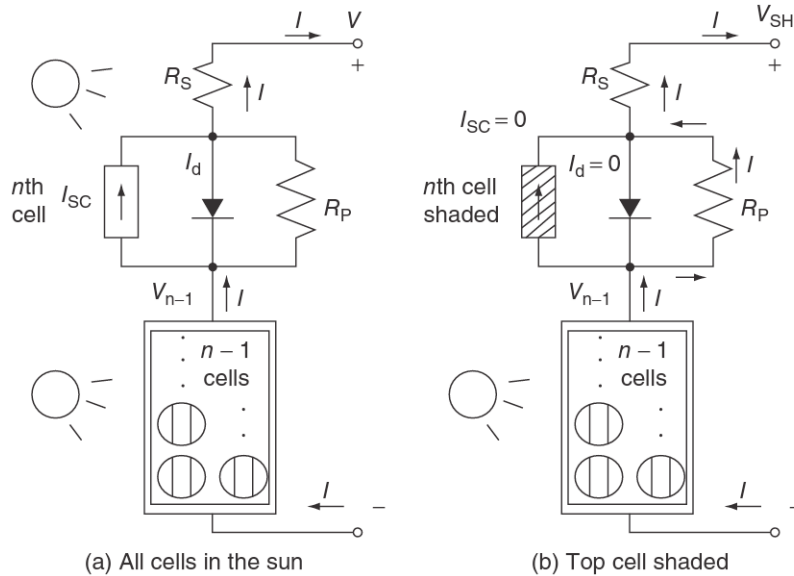
At $S = 0.8 \text{ kW/m}^2$:

$$T_{\text{cell}} = T_{\text{amb}} + 25^\circ\text{C}$$

which matches the NOCT definition condition.

Shading Impact: Voltage Drop

Even a small fraction of a single module is shaded, the performance of an entire string of modules can be compromised.



$$V_{SH} = V_{n-1} - I(R_P + R_S)$$

$$= \left(\frac{n-1}{n} \right) V - I(R_P + R_S)$$

$$\Delta V = V - V_{SH} = V - \left(1 - \frac{1}{n} \right) V + I(R_P + R_S)$$

$$= \frac{V}{n} + I(R_P + R_S) \quad \text{Equation 5.29}$$

$$\cong \frac{V}{n} + I \cdot R_P \quad (R_P \gg R_S)$$

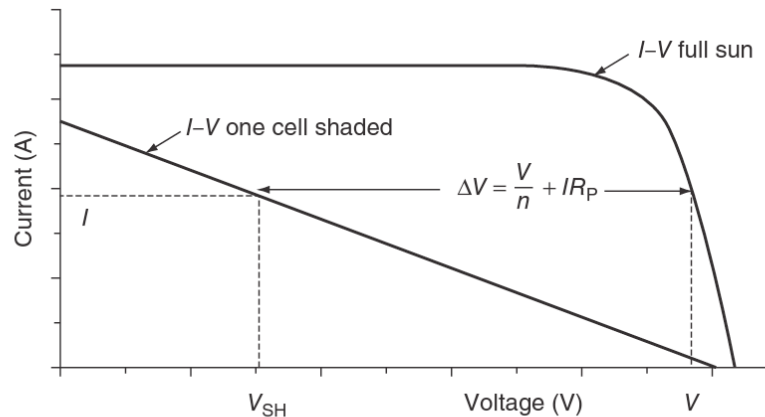


FIGURE 5.44 Effect of shading one cell in an n-cell module. At any given current, module voltage drops from V to $V - \Delta V$.

Example

Example 5.6 Impacts of Shading on a PV Module. The 72-cell PV module described in Example 5.4 had a parallel resistance per cell of $R_P = 10.0 \, \Omega$ and a series resistance R_S of $0.001 \, \Omega$. In full sun and at current $I = 5.73 \, \text{A}$, the output voltage of the module was found to be $V = 40.63 \, \text{V}$. If one cell is shaded, find the following if somehow the same $5.73 \, \text{A}$ is forced to flow through the module.

- The module output voltage.
- Power dissipated in the shaded cell.

Solution

- From Equation 5.29, the drop in module voltage will be

$$\Delta V = \frac{V}{n} + I(R_P + R_S) = \frac{40.63}{72} + 5.73 \times (10.0 + 0.001) = 57.87 \, \text{V}$$

(obviously, we could have ignored R_S).

The module output voltage will now be $40.63 - 57.87 = -17.24 \, \text{V}$

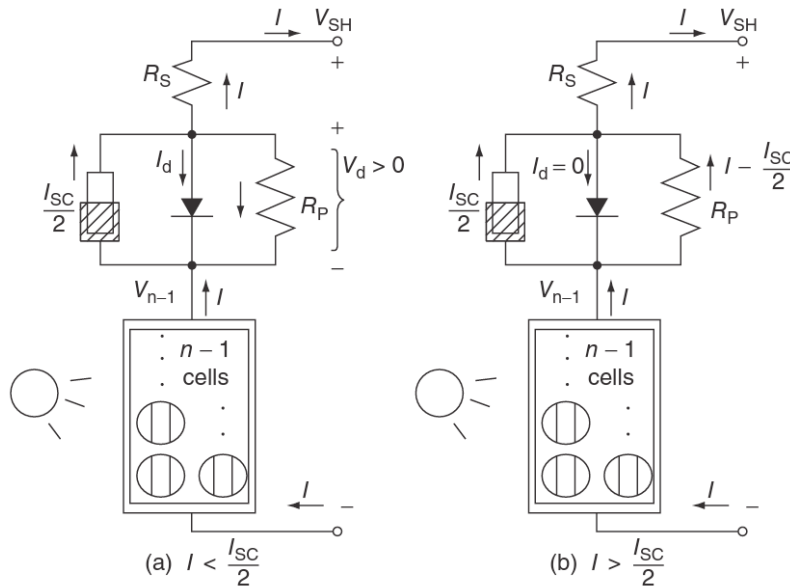
(Yes, this is possible. For example, if this module with one shaded cell is part of a string of modules and the good ones are trying to drive $5.73 \, \text{A}$ through the whole string, then it would be better to completely remove that shaded module from the string than to leave it in place!)

- Since that $5.73 \, \text{A}$ flows through both R_S and R_P , the power dissipated in that shaded cell will be

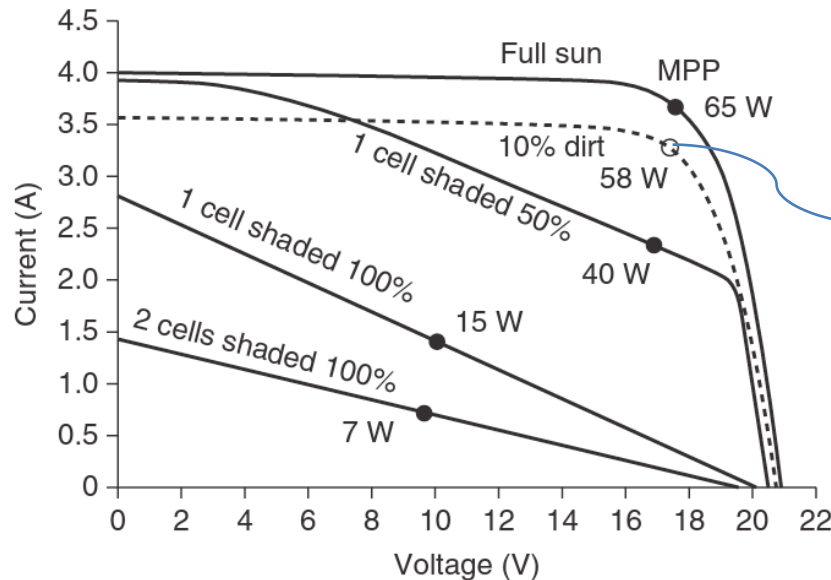
$$P = I^2(R_P + R_S) = (5.73)^2(10.0 + 0.001) = 69.7 \, \text{W}$$

All of that power dissipated in the shaded cell is converted to heat, which can cause a local hot spot that may permanently damage the plastic laminates enclosing the cell.

Partial Shading Impacts on I-V Curves



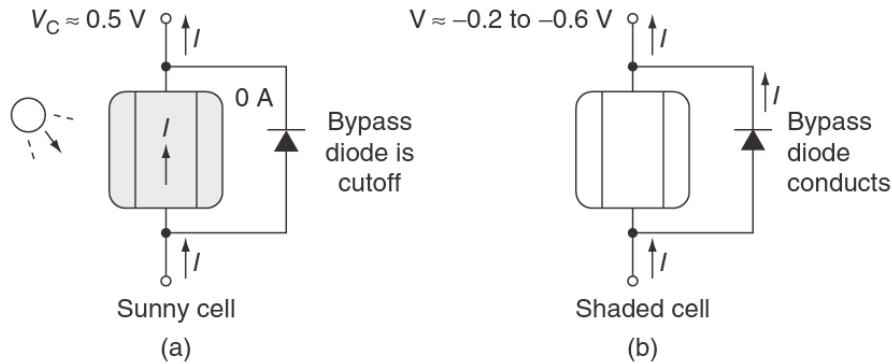
- (a) if $I < I_{SC}/2$, the diode still conducts with 1 cell half-shaded
- (b) if $I > I_{SC}/2$, the diode shuts off, and current passes through R_P in an opposite direction which can result in large voltage drop in V_{SH} .



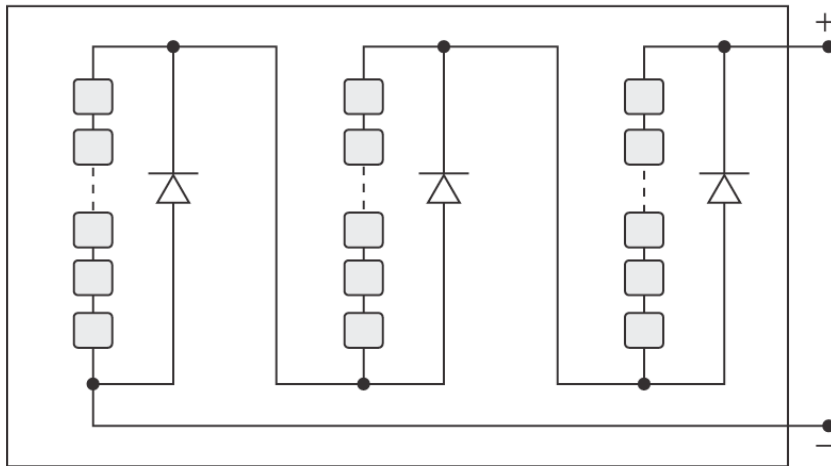
Uniformly distributed dirt.

When modules get dirty, the I-V curve shifts downward just the way it would if the Sun were a bit less intense.

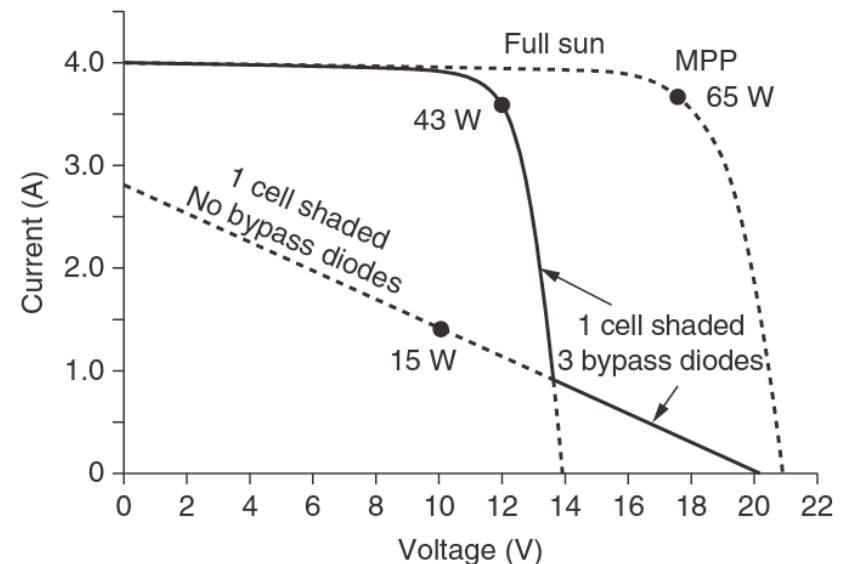
Shade Mitigation: Bypass Diodes



- (a) Bypass diode is cut off due to the voltage rise across the cell in the Sun
- (b) When the cell is shaded, the current is diverted to flow through the diode, which limits the voltage drop to $0.2 \sim 0.6 \text{ V}$, rather than a large drop if there is no bypass diode.



3 bypass diodes, each covering 1/3 of the string



Bypass Diodes (cont'd)

Bypass diodes help current go around a shaded/malfunctioning module within a string. This

1. improves the string performance
2. prevents hot spots from developing in individual shaded cells

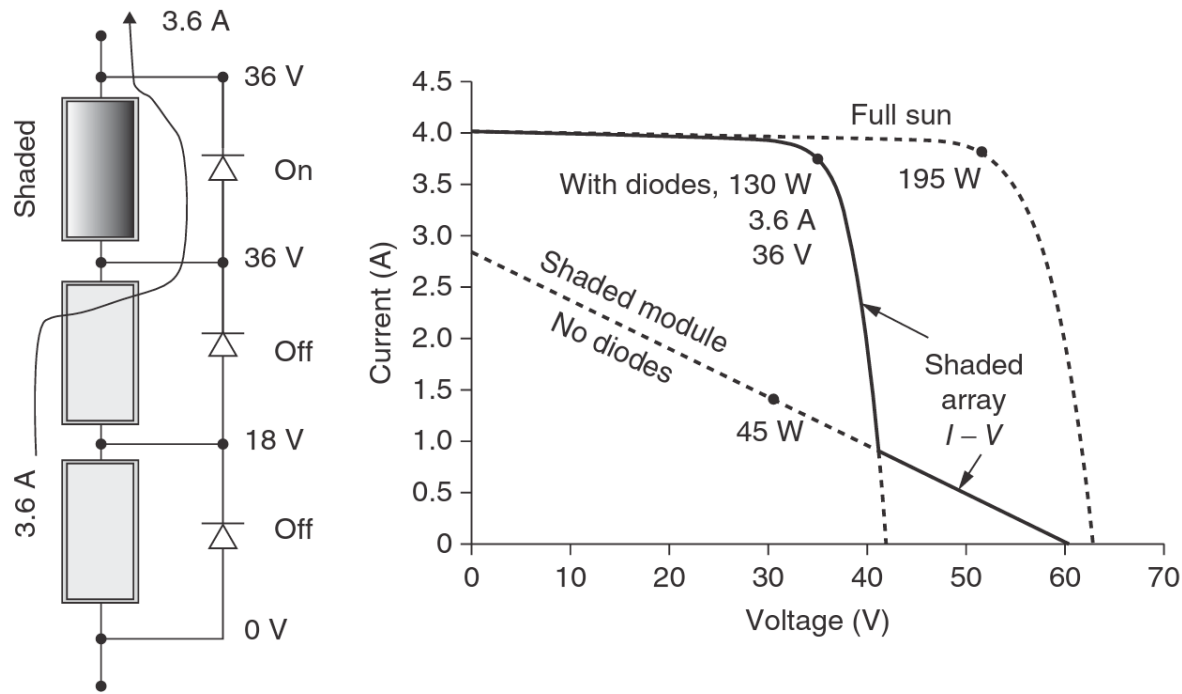
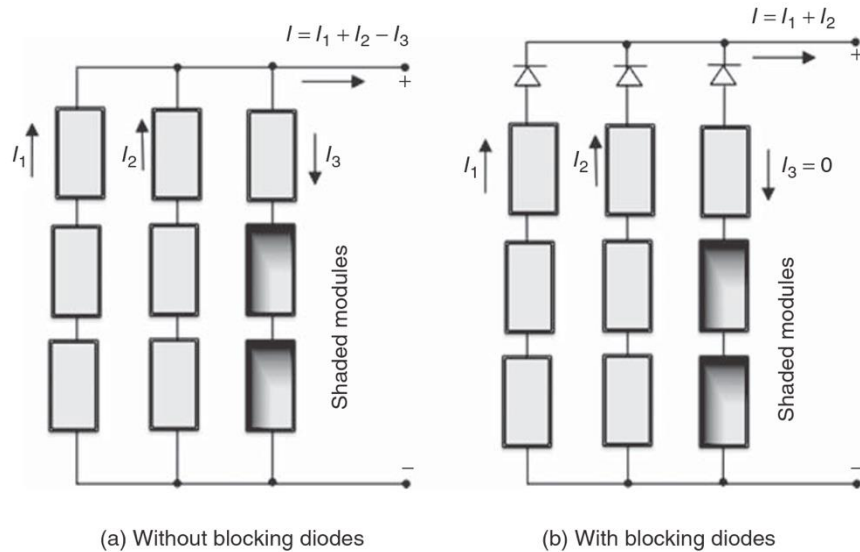


FIGURE 5.50 Showing the ability of bypass diodes to mitigate shading problems in a string of modules.

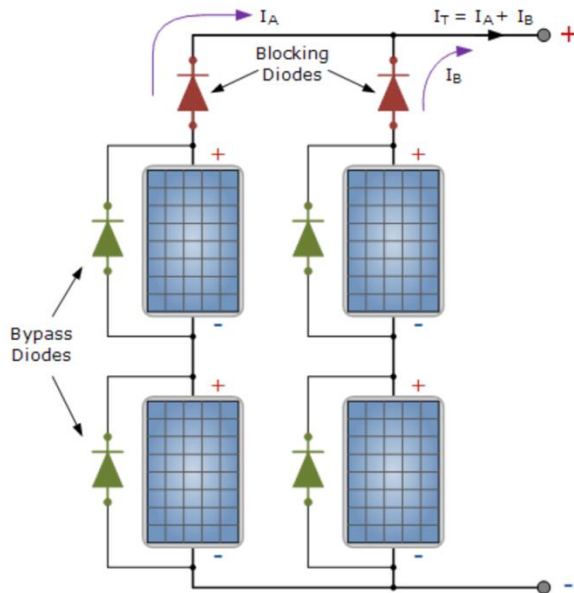
Shade Mitigation: Blocking Diodes



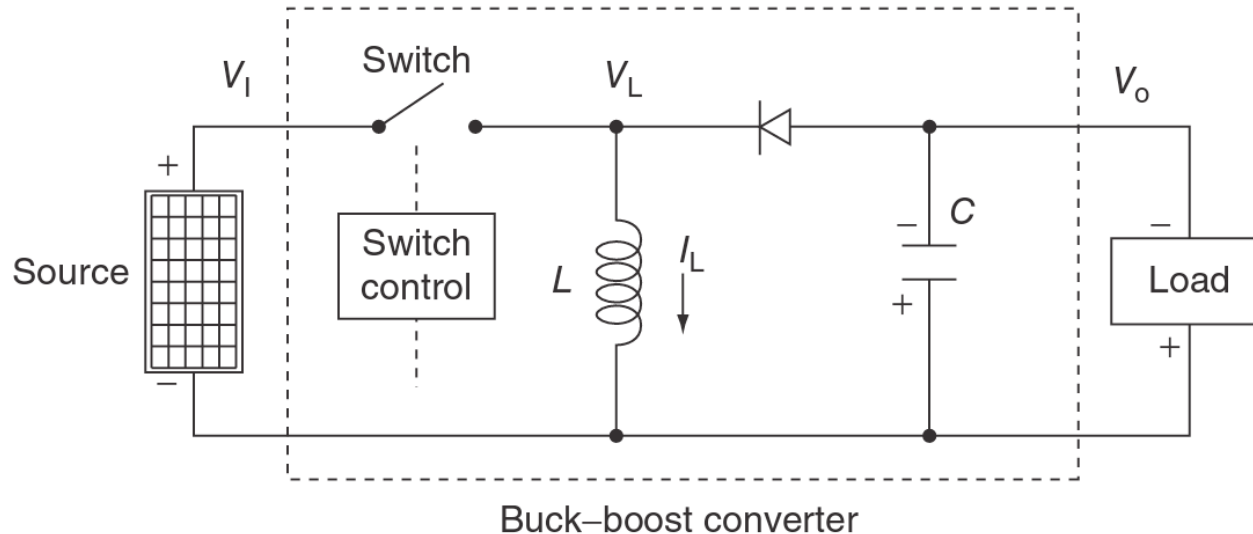
A malfunctioning or shaded string can **withdraw current** from the rest of the array.

Adding blocking diodes at the top of each string can prevent

- the current generated by the other parallel connected PV panels in the same array flowing back through a weaker (shaded) network
- the fully charged batteries from discharging or draining back through the array at night.



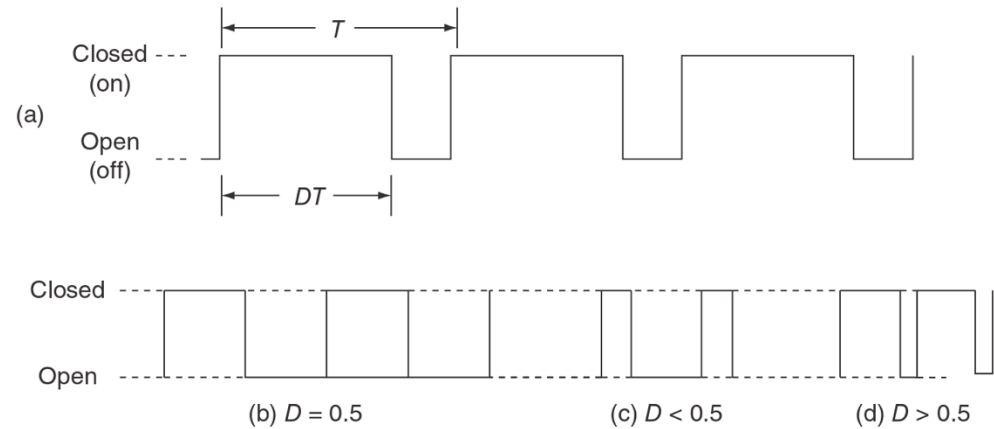
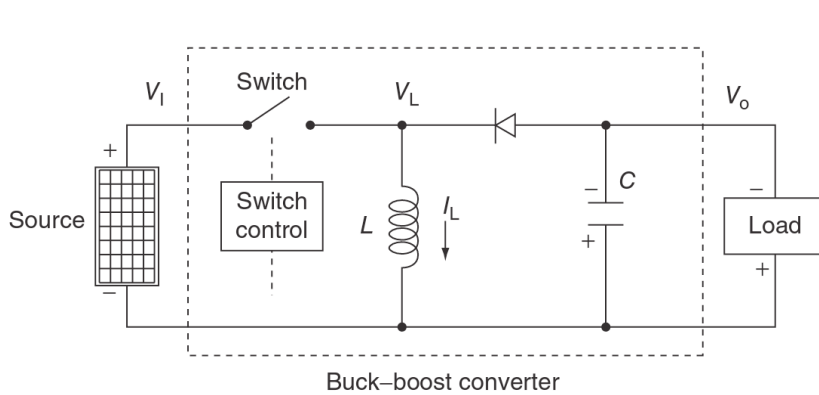
Buck-Boost Converter



A buck-boost converter is a DC-DC power electronic converter that can

- Step down (buck) the voltage
- Step up (boost) the voltage
- The transistor switch flips on and off at a rapid rate (on the order of 20 kHz) under some sensing/controlling signals and logic algorithms.
- Insight: with fast switching, I_L and V_O are essentially constant.

Duty Cycle

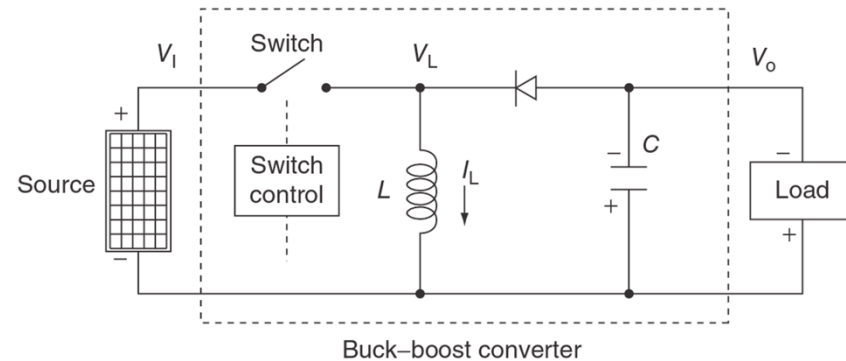


$$\text{Duty cycle} = \frac{DT}{T}$$

Which is the fraction of the time when the switch is closed

Change DC Voltage Levels

Analysis is based on the energy balance for the magnetic field of the inductor:

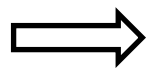


The average power put into the magnetic field of the inductor:

$$\bar{P}_{L, \text{ in}} = \frac{1}{T} \int_0^{DT} V_i I_L dt = V_i I_L D$$

When the switch opens, the magnetic field begins to collapse, and starting supporting the load. The average power delivered:

$$\bar{P}_{L, \text{ out}} = \frac{1}{T} \int_{DT}^T V_L I_L dt = \frac{1}{T} \int_{DT}^T V_o I_L dt = V_o I_L (1 - D)$$



$$\frac{V_o}{V_i} = - \left(\frac{D}{1 - D} \right)$$

The negative sign denotes that the reference voltage direction of the load is opposite to the one of the source, as shown in the figure above

$$D = \frac{1}{2} \rightarrow V_o = V_i, \quad D = \frac{2}{3} \rightarrow V_o = 2V_i, \quad D = \frac{1}{3} \rightarrow V_o = \frac{1}{2}V_i.$$

Example

Example 5.7 Duty Cycle for a MPPT. Under certain ambient conditions, a particular PV module has its maximum power point at $V_m = 30$ V and $I_m = 6$ A. What duty cycle should be provided to a buck–boost converter if the module is to deliver 12 V to charge a battery? How many amperes would be delivered to the battery? If the ambient were to cool off some without a change in insolation, should the duty cycle be increased or decreased?

Solution.
$$\frac{12}{30} = \left(\frac{D}{1 - D} \right) = 0.4 \quad \Longrightarrow \quad D = \frac{0.4}{1.4} = 0.286$$

Since we will assume a 100% efficient converter, input power equals output power so that

$$V_{PV} \cdot I_{PV} = V_{\text{Battery}} \cdot I_{\text{Battery}}$$

$$I_{\text{Battery}} = \frac{30\text{V} \cdot 6\text{A}}{12\text{V}} = 15\text{A}$$

With cooler temperatures, the PV voltage at the MPP increases somewhat (e.g., see Fig. 5.35) so the duty cycle D should be decreased a bit.

$$D = \frac{V_o}{V_o + V_i} \quad \Longrightarrow \quad \text{increased } V_i \text{ yields decreased } D$$

Tracking Maximum Power Point (MPP)

The key challenge is:

V_{mpp} is *NOT* constant

It changes continuously with:

- solar irradiance
- cell temperature
- partial shading
- aging
- panel mismatch

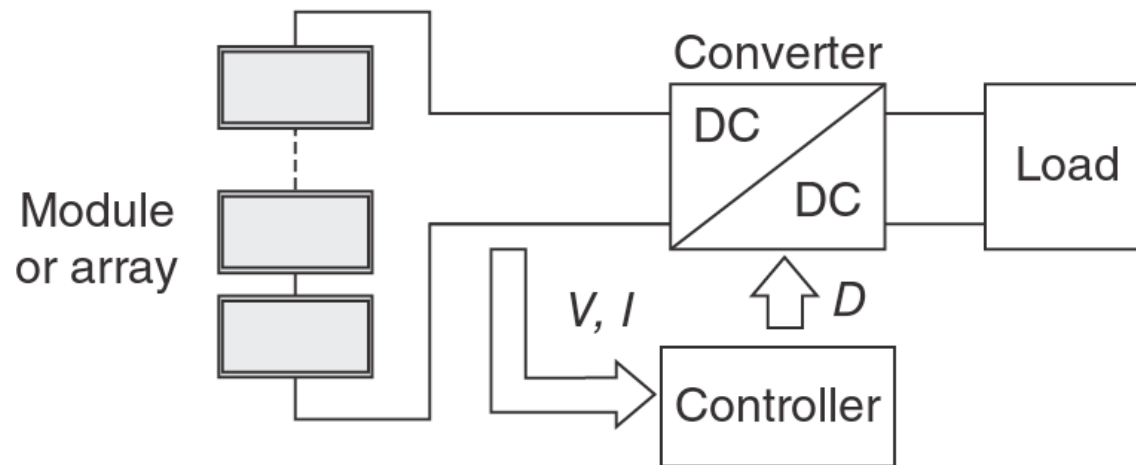
Therefore, the PV system usually does **not know the true current MPP location ahead of time.**

Under changing environmental conditions, the panel moves among many different I-V curves. Each curve has a different: V_{mpp} , I_{mpp} , P_{max}

So the controller must continuously:

- 1) observe the operating condition
- 2) estimate where the new MPP is
- 3) adjust converter duty cycle
- 4) track the moving optimum.

MPPT Control via Duty Cycle



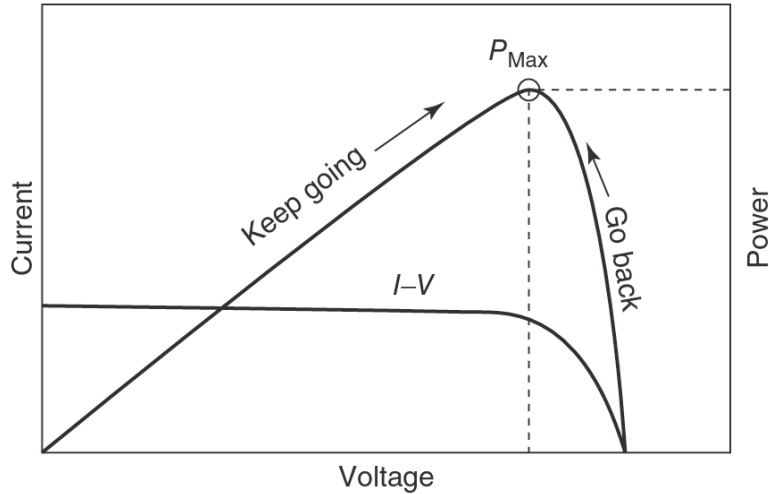
The converter does **not directly command panel voltage perfectly**. Instead, it changes:

- duty cycle D
- effective load impedance seen by the PV panel

which indirectly changes:

- PV voltage
- PV current
- The MPPT algorithm determines how to adjust D .

Incremental Conductance (IncCond) MPP Method



$$\frac{dP}{dV} = I \frac{dV}{dV} + V \frac{dI}{dV} = I + V \frac{dI}{dV} \approx I + V \frac{\Delta I}{\Delta V}$$

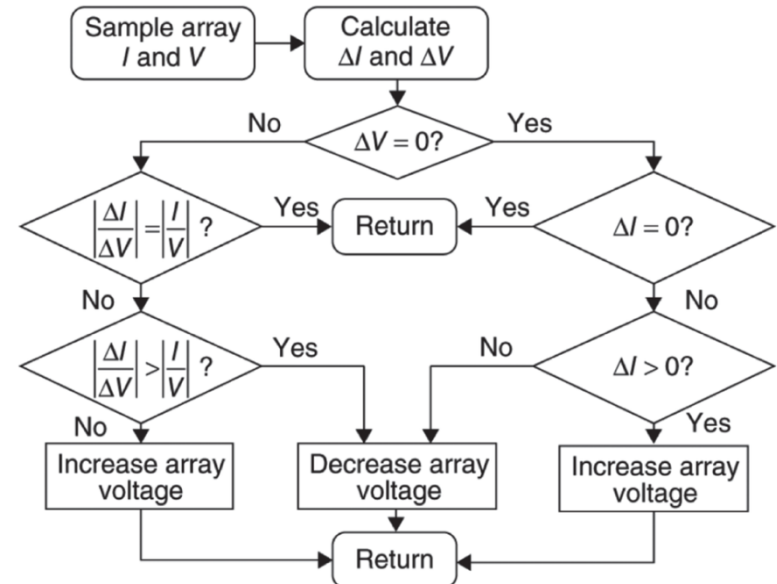
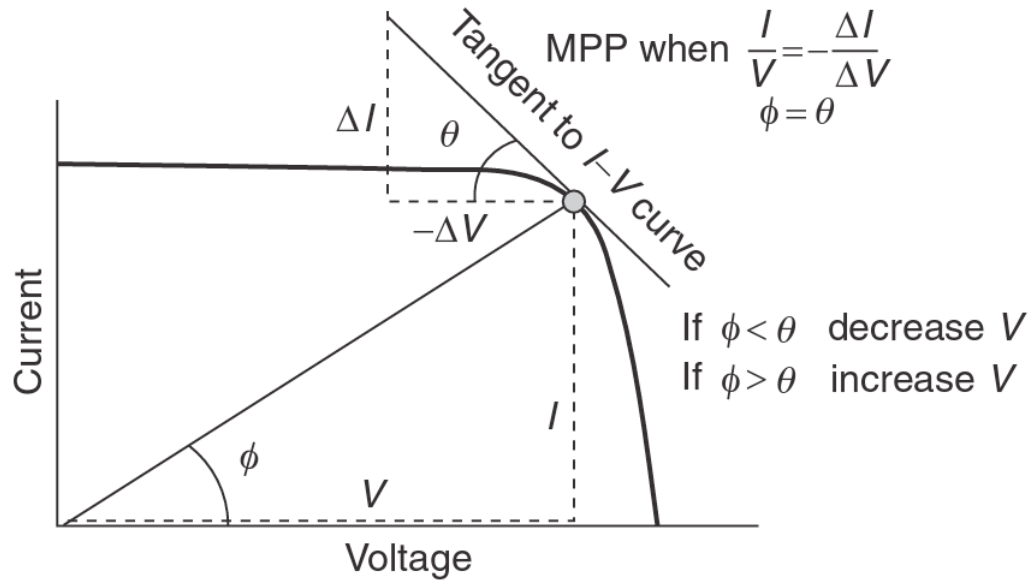
At the MPP: $\frac{dP}{dV} = 0 \quad \Rightarrow$

$$\frac{\Delta I}{\Delta V} = -\frac{I}{V}$$

incremental
conductance

instantaneous
conductance

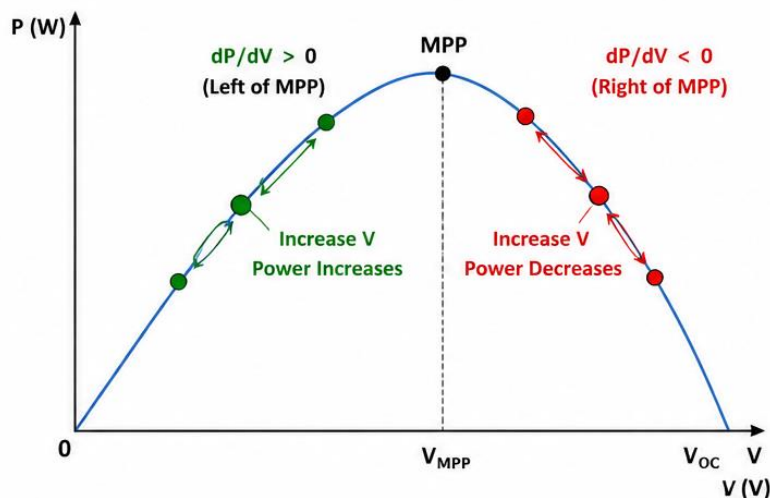
IncCond Algorithm Flowchart



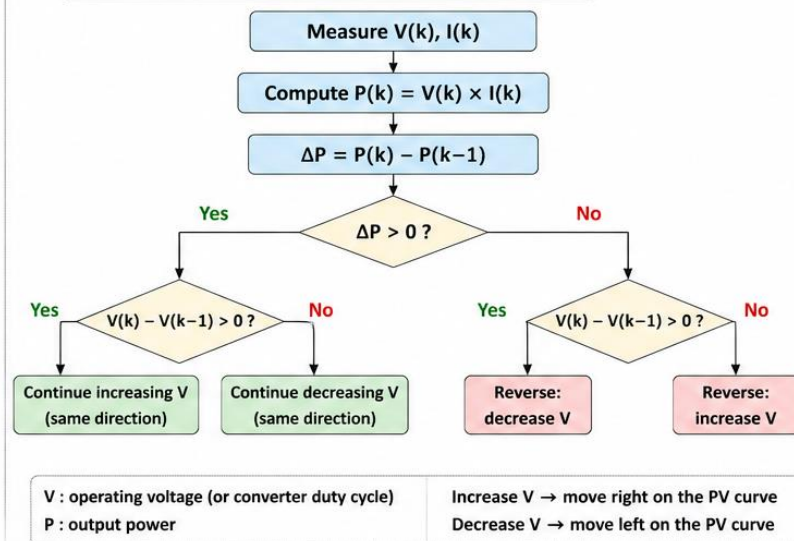
Perturb and Observe (P&O) algorithm

P&O algorithm: the most widely used MPPT method in photovoltaic systems because of its simplicity and low implementation cost.

1) Basic Idea



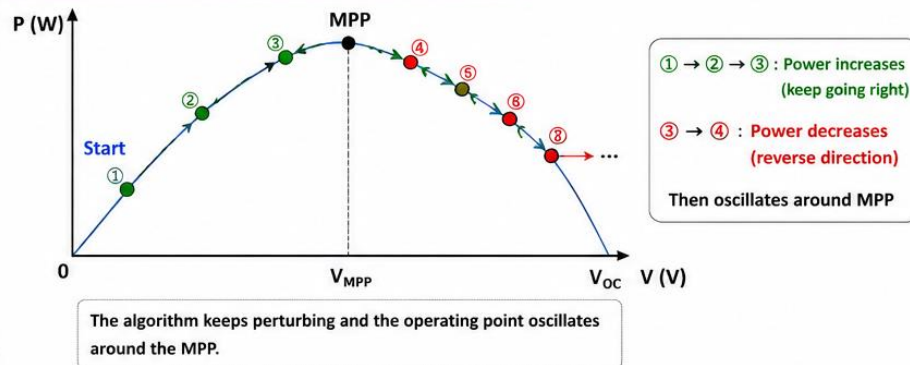
2) P&O Algorithm (at each sampling instant)



3) Decision Table

ΔP ($P(k) - P(k-1)$)	ΔV ($V(k) - V(k-1)$)	Action	Meaning
> 0	> 0	Increase V	Moved toward MPP
> 0	< 0	Decrease V	Moved toward MPP
< 0	> 0	Decrease V	Moved away from MPP
< 0	< 0	Increase V	Moved away from MPP

4) How the Operating Point Moves (Example)



Algorithm Comparison: IncCond vs P&O

Algorithm	Pros	Cons
Perturb & Observe (P&O)	Simple, low-cost, easy to implement, widely used	Oscillates around MPP and performs poorly under rapidly changing irradiance
Incremental Conductance (IncCond)	More accurate and faster under changing weather conditions	More computationally complex and requires more sensing/calculation